

DNA/Protein Sequence Analysis using BLAST

Introduction

Bioinformatics is an interdisciplinary field that combines biology and computational tools to analyze biological data. One of the most widely used bioinformatics tools is the Basic Local Alignment Search Tool (BLAST), developed by the National Center for Biotechnology Information (NCBI). BLAST enables researchers to compare DNA or protein sequences with existing database sequences, identify homologous proteins, and study evolutionary relationships.

In this analysis, the Human Insulin protein sequence was retrieved from the UniProt database and analyzed using the NCBI Protein BLAST (BLASTP) tool to identify similar protein sequences and evaluate sequence conservation among different organisms.

Objective

The objective of this study is to analyze the Human Insulin protein sequence using the NCBI Protein BLAST tool and identify homologous protein sequences by comparing them with sequences available in the NCBI protein database.

Materials and Software Used

- *UniProt Protein Database*
- *NCBI Protein BLAST (BLASTP)*
- *Internet Browser*

Methodology

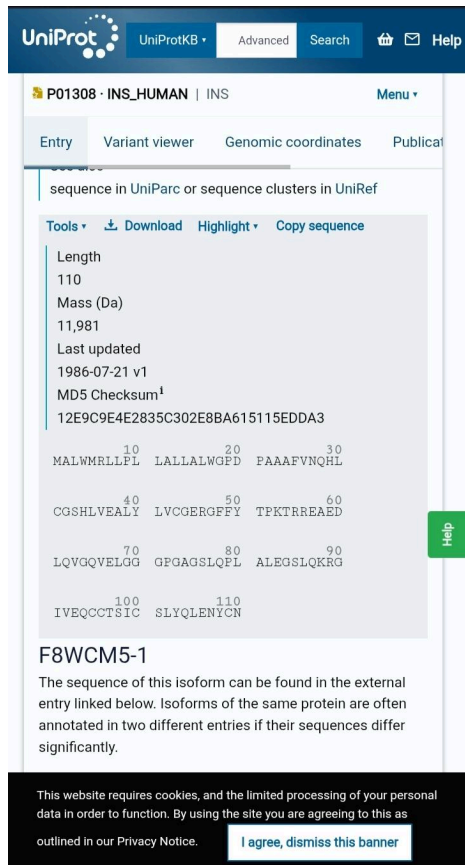
The Human Insulin protein sequence was obtained from the UniProt database. The retrieved protein sequence was copied and submitted to the NCBI Protein BLAST (BLASTP) program using the default search parameters. The BLAST search compared the query sequence with protein sequences available in the NCBI database. The resulting alignments were examined based on query coverage, percentage identity, bit score, and E-value to determine the most significant homologous protein sequence.

Results and Discussion

The BLAST analysis successfully identified multiple homologous insulin protein sequences from different organisms. The highest-scoring match corresponded to the insulin preproprotein of Pan troglodytes (Chimpanzee). The alignment showed 98% sequence identity and 99% positive matches, with an E-value of $1e-72$ and a Bit Score of 221, indicating an extremely significant sequence similarity. These results demonstrate that the insulin protein is highly conserved among closely related species and emphasize the importance of BLAST in comparative sequence analysis and evolutionary studies.

Screenshots

☐ **UniProt Protein Sequence**



The screenshot displays the UniProt protein sequence page for P01308 (INS_HUMAN). The page includes a navigation bar with 'UniProtKB', 'Advanced', 'Search', and 'Help' links. The main content area shows the protein name 'P01308 · INS_HUMAN | INS' and a 'Menu' dropdown. Below this, there are tabs for 'Entry', 'Variant viewer', 'Genomic coordinates', and 'Publications'. The 'Entry' tab is selected, showing the protein sequence in UniProt or sequence clusters in UniRef. The sequence is displayed in a light blue box with line numbers 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, and 110. The sequence is: MALWMRLRLPL LALLALWGPD FAAAFVFNQHL CGSHLVEALY LVCGERGFFY TPKTRREAED LQVGQVELGG GPGAGSLQPL ALEGLQKRG IVEQCCTSIIC SLYQLENYCN. A 'Tools' dropdown menu is visible, with options for 'Download', 'Highlight', and 'Copy sequence'. A green 'Help' button is located on the right side of the sequence box. Below the sequence, the isoform 'F8WCM5-1' is mentioned, with a note that the sequence of this isoform can be found in the external entry linked below. At the bottom, there is a cookie consent banner.

UniProtKB Advanced Search Help

P01308 · INS_HUMAN | INS Menu

Entry Variant viewer Genomic coordinates Publications

sequence in UniProt or sequence clusters in UniRef

Tools Download Highlight Copy sequence

Length
110
Mass (Da)
11,981
Last updated
1986-07-21 v1
MD5 Checksum¹
12E9C9E4E2835C302E8BA615115EDDA3

10 20 30
MALWMRLRLPL LALLALWGPD FAAAFVFNQHL

40 50 60
CGSHLVEALY LVCGERGFFY TPKTRREAED

70 80 90
LQVGQVELGG GPGAGSLQPL ALEGLQKRG

100 110
IVEQCCTSIIC SLYQLENYCN

F8WCM5-1

The sequence of this isoform can be found in the external entry linked below. Isoforms of the same protein are often annotated in two different entries if their sequences differ significantly.

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Human Insulin protein sequence retrieved from the UniProt database

☐ **BLAST Result Summary**

BLAST Result Table

10:09

0.20 5G+ 47%

VA LTE KHz

m.nih.gov

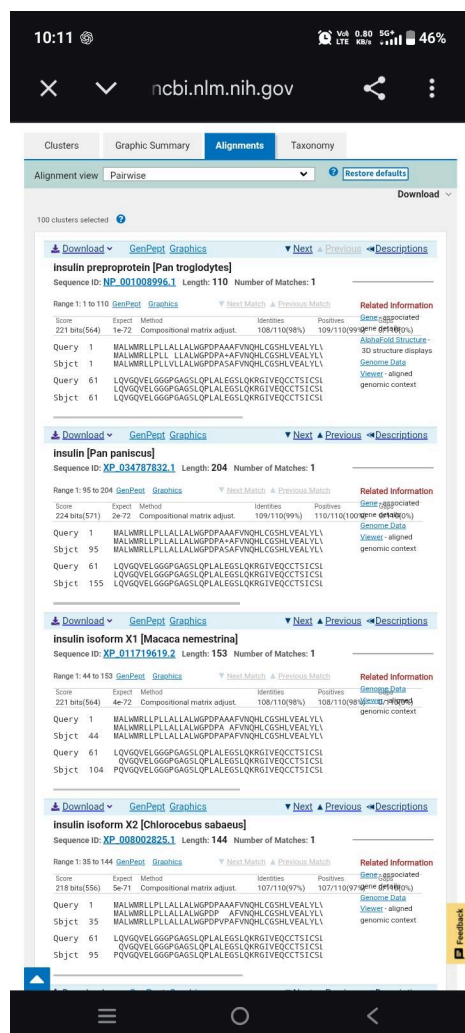
GapPeet Graphics Distance tree of results Multiple alignment MSA Viewer

Cluster Composition	Cluster	Cluster	Max Score	Total Score	Query Cover	E value	Per. ident	Acc. Len	Accession
Click the icon to see the cluster contents									
16 member(s), 12 organism(s)	elaces - insulin.oc	221	221	100%	1e-72	88.18%	110	NP_001008996.1	
3 member(s), 3 organism(s)	anmat - insulin.RP	224	224	100%	2e-72	99.09%	204	XP_034707832.3	
5 member(s), 5 organism(s)	anmat - insulin.la	221	221	100%	4e-72	88.18%	153	XP_011717619.2	
1 member(s), 1 organism(s)	anmat - insulin.la	218	218	100%	5e-71	97.27%	144	XP_008002825.3	
1 member(s), 1 organism(s)	black.s - insulin.la	218	218	100%	1e-70	97.27%	147	XP_017743290.3	
1 member(s), 1 organism(s)	anmat - insulin.la	218	218	100%	7e-70	97.27%	209	XP_008002752.3	
5 member(s), 5 organism(s)	anmat - INS isofo	191	191	100%	1e-60	88.18%	98	PN_116433.1	
2 member(s), 2 organism(s)	elaces - insulin.Tr	184	184	100%	4e-58	90.00%	110	KAN7991649.1	
1 member(s), 1 organism(s)	Europe - insulin.lm	184	184	100%	5e-58	79.09%	110	XP_080768357.2	
1 member(s), 1 organism(s)	other.s - ZZ.coala	183	183	78%	3e-56	100.00%	210	UE92608.1	
4 member(s), 4 organism(s)	hals - insulin.RP	178	178	100%	2e-55	88.18%	110	KAN3685427.1	
1 member(s), 1 organism(s)	hah.G - brnetho	179	179	78%	2e-55	100.00%	143	MGZ4612290.1	
2 member(s), 1 organism(s)	other.s - HPII.fon	175	175	78%	1e-54	98.84%	97	AGG11849.1	
1 member(s), 1 organism(s)	Rhesu - insulin.oc	174	174	78%	3e-54	98.84%	86	1006230A	
1 member(s), 1 organism(s)	red.vo - insulin.la	175	175	100%	5e-54	85.45%	128	XP_049922438.1	
1 member(s), 1 organism(s)	Alabof - brnetho	175	175	100%	9e-54	85.45%	145	KAL6031000.1	
1 member(s), 1 organism(s)	other.s - SUMD.L	176	176	78%	1e-53	97.67%	207	UHG24472.1	
3 member(s), 2 organism(s)	alabita - insulin.oc	173	173	100%	2e-53	86.36%	110	NP_001353985.1	
1 member(s), 1 organism(s)	cosper - insulin.RP	175	175	100%	2e-53	87.27%	184	KAL0628984.1	
1 member(s), 1 organism(s)	E.coli - brnetho	172	172	95%	3e-53	84.76%	109	HDJ054968.1	
1 member(s), 1 organism(s)	Chines - insulin.lc	177	177	100%	3e-53	86.36%	248	EGW08477.1	
1 member(s), 1 organism(s)	Back.v - brnetho	172	172	100%	6e-53	84.55%	155	KAN7801657.1	
2 member(s), 2 organism(s)	New.W - insulin.la	173	173	100%	6e-53	86.36%	155	XP_064276392.2	
2 member(s), 2 organism(s)	anmat - insulin.la	171	171	100%	8e-53	84.55%	110	XP_003798420.1	
4 member(s), 4 organism(s)	New.W - insulin.la	171	171	100%	9e-53	85.45%	108	BS7923.1	
9 member(s), 7 organism(s)	gost - insulin.L	171	171	100%	1e-52	82.73%	110	NP_0636092.1	
1 member(s), 1 organism(s)	other.s - hanaLuc	177	177	100%	2e-52	81.82%	392	XON88723.1	
1 member(s), 1 organism(s)	black.B - insulin.RP	170	170	100%	3e-52	84.55%	110	XP_006911099.9	
10 member(s), 10 organism(s)	elaces - insulin.oc	170	170	100%	3e-52	88.18%	110	NP_001123565.1	
16 member(s), 29 organism(s)	rod - insulin.2	169	169	100%	4e-52	81.82%	110	NP_001172012.1	
4 member(s), 4 organism(s)	anmat - insulin.RP	169	169	100%	4e-52	87.27%	109	XP_063527944.1	
1 member(s), 1 organism(s)	red.fox - insulin.s	169	169	100%	6e-52	88.18%	124	XP_072614830.3	
1 member(s), 1 organism(s)	North - insulin.RP	168	168	100%	2e-51	80.91%	110	KAN6168390.1	
1 member(s), 1 organism(s)	white - insulin.s	169	169	100%	2e-51	85.45%	148	XP_035119677.2	
1 member(s), 1 organism(s)	stined - insulin.lh	167	167	100%	4e-51	83.64%	110	XP_039001000.1	
1 member(s), 1 organism(s)	grey.w - insulin.lc	169	169	100%	5e-51	88.18%	177	MEW01439.1	
1 member(s), 1 organism(s)	other.s - insulin.L	166	166	77%	8e-51	94.39%	105	AWG14628.1	
4 member(s), 4 organism(s)	rodenda - insulin.L	166	166	100%	1e-50	78.18%	110	XP_052028151.1	
1 member(s), 1 organism(s)	hamed - insulin.la	167	167	100%	2e-50	81.82%	171	GAR5527225.1	
3 member(s), 2 organism(s)	gethla - RecNano	165	165	100%	2e-50	80.91%	110	G62587.1	
1 member(s), 1 organism(s)	North - LOW.GU	169	169	100%	2e-50	88.18%	234	XP_061089293.1	
5 member(s), 4 organism(s)	Old.Vo - insulin.la	164	164	100%	9e-50	85.45%	135	XP_025211012.2	
6 member(s), 6 organism(s)	cat.la - insulin.oc	163	163	100%	1e-49	80.91%	110	NP_001009272.2	
1 member(s), 1 organism(s)	gray.se - insulin.lh	164	164	100%	2e-49	85.45%	148	XP_035930639.2	
1 member(s), 1 organism(s)	gnater - insulin.RP	162	162	100%	4e-49	89.09%	123	XP_032975462	
1 member(s), 1 organism(s)	overkat - insulin.la	161	161	100%	8e-49	86.36%	110	XP_029770545	
9 member(s), 9 organism(s)	rodenda - insulin.oc	161	161	100%	9e-49	89.09%	110	NP_001269188	
1 member(s), 1 organism(s)	naked - insulin.lh	160	160	100%	1e-48	79.09%	110	XP_004802148	
1 member(s), 1 organism(s)	other.s - SEAP.E2	173	173	100%	2e-48	81.82%	652	WVG21740.1	

Feedback

Homologous protein sequences identified by NCBI BLAST.

BLAST Alignment



Pairwise sequence alignment of Human Insulin with the top homologous protein

Conclusion

The Human Insulin protein sequence was successfully analyzed using the NCBI Protein BLAST tool. The analysis identified highly homologous protein sequences with excellent alignment statistics, confirming the evolutionary conservation of the insulin protein. This study demonstrates the effectiveness of BLAST as a reliable tool for sequence similarity analysis, protein identification, and comparative bioinformatics research.

References

- 1.UniProt Consortium. UniProt Knowledgebase (UniProtKB). <https://www.uniprot.org/>
- 2.National Center for Biotechnology Information (NCBI). Basic Local Alignment Search Tool (BLAST).

<https://blast.ncbi.nlm.nih.gov/>